UEI518: ARTIFICIAL INTELLIGENCE AND APPLICATIONS

L	T	P	Cr
3	1	2	4.5

Course Objectives: The student should study the concepts of artificial intelligence and learn the methods of solving problems using artificial intelligence.

Overview: Definition, scope, foundations, approaches, and applications of AI; AI: past, present, and future.

Agents and Environments: agents; rationality; types of agents; properties of environments.

State Space Representation: State and operators; state space; representation real world problems as state space, problem characteristics.

Searching Strategies: uninformed searching methods (DFS, BFS, DFS-ID); informed searching methods such as best first search, hill climbing, A*, iterative deepening A*; problem reduction; constraint satisfaction problems; neural, stochastic, and evolutionary algorithms

Game Playing: Game theory and optimal decisions; Turn-taking games; Adversarial search; Minimax principle; Monte-Carlo tree search; Alpha-Beta pruning.

Dealing with uncertainty: probability, connection to logic, independence, Bayes rule, Bayesian networks, probabilistic inference; time and uncertainty, hidden Markov model; Decision making-Utility theory, utility functions, Decision theoretic expert systems

Fuzzy Systems: Fuzzy sets, Operation on fuzzy sets, Fuzzy relations, Membership functions, fuzzy to crisp conversions, Fuzzy reasoning, Fuzzy Inference Systems, Design of Fuzzy logic controller, Advantages of fuzzy logic controller over a conventional PID controller

Neural Network as Learning Machine: Mathematical model of neuron, activation functions, types of learning, learning methods, classification of neural networks, perceptron and multilayer perceptron, gradient and error back-propagation learning algorithms, typical applications of feedforward and feedback neural network, Neuro-Fuzzy hybrid system; Engineering Applications

Optimization with Genetic Algorithms:

Evolutionary algorithms, Genetic Algorithms, Genetic Fuzzy Systems, Genetic Neural Systems

Course Learning Outcomes (CLO): At the end of the course, the student should be able to:

1. Analyze appropriate AI methods to solve a given problem that are amenable to solution by AI.
2. Formalize a given problem in the language/framework of different AI methods.
3. Implement Fuzzy theory for engineering applications.

4. Implement neural network as learning machine for engineering applications.

Text Books:

1. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill- 2008. (Units- I,II,VI & V).

2. Dan W. Patterson, "Introduction to AI and ES", Pearson Education, 2007. (Unit-III).

3. Ross, J. T., Fuzzy Logic with Engineering Applications, McGraw Hill (1995).

4. S. Haykin, Neural Network: A Comprehensive Foundation, Pearson Education (2003).

References Books

1. S.N. Sivanandam, S.N. Deepa "Soft Computing", 2nd Edition, Wiley India, 2012.

2. Stuart Russel and Peter Norvig "AI – A Modern Approach", 2nd Edition, Pearson Education 2007.

3. Deepak Khemani "Artificial Intelligence", Tata Mc Graw Hill Education 2013.

Evaluation Scheme:

S.No.	**Evaluation Elements**	**Weightage (%)**
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes/Lab Evaluations/Project)	30